Tidal: Direction-Paired Virtual-Channel Buffer Pooling with a 1R1W Shared Pool

Changxun Pan Liming Wang
Equal contribution: 50% each

Abstract

Conventional on-chip routers bind virtual-channel (VC) storage to individual input directions. This organization strands storage when one direction is congested and its opposite peer is idle. TIDAL reorganizes the same eight VC slots per opposing-direction pair into two two-slot reserved regions and a four-slot shared POOL. Each direction retains one private adaptive VC and one private escape VC. The POOL provides only one read and one write service per cycle; alternating read phases let one side inspect its reserved region while the other inspects the POOL. Four conserved POOL credits control admission. A pressure policy moves released credits toward the side with fuller reserved storage, keeps the previous owner on ties, caps ownership at three, and reclaims only idle peer-owned slots on demand.

We implement the design in gem5 Garnet and evaluate an equal-storage four-private-VC baseline. Across 1,077 main-matrix simulations, all runs completed without deadlock and recorded zero cross-direction POOL read conflicts. Under persistent 90/10 X-direction skew with eight-cycle links, TIDAL lowers network latency by 9.2% and extends the stable offered rate from 0.08 to 0.10. At a fully skewed point beyond baseline saturation it delivers 1.818 times as many packets. Under tornado traffic, pressure-aware credit control lowers latency by 4.85%, whereas blind round-robin ownership loses 56.5% throughput. Balanced, short-link, and rapidly reversing traffic can be slower. The results identify persistent directional skew near an input credit limit as the useful operating region and motivate timing and RTL cost validation.

Keywords: on-chip networks, virtual channels, shared buffers, credit flow control, Garnet, gem5

1 Introduction

Virtual-channel (VC) flow control improves throughput by multiplexing logical channels over physical links, and it turns buffer organization into a central router design choice: input buffers account for a large share of an on-chip router's area and power budget [5, 10]. Garnet models a pipelined, credit-controlled network inside gem5 and is widely used to study these tradeoffs [1, 3]. Its conventional organization is fully private: every input direction owns a fixed set of VCs, and storage never migrates.

Private storage is wasteful precisely when the network is most stressed. Directionally skewed traffic—tornado permutations, biased dimension-ordered flows—drives one member of an opposing-direction pair against its input-credit limit while the peer sits nearly empty. The cost is measurable: at eight-cycle link latency, a fully skewed workload saturates a private four-VC baseline at an offered rate of 0.08, whereas the same eight buffer slots per pair, reorganized by TIDAL, sustain 0.10 and deliver 1.818 times as many packets beyond the baseline's saturation point (Section 6.3).

Elastic buffers are the classic remedy, but prior schemes buy their elasticity with multiported storage or wide arbitration, and industrial credit pools share only among the message classes of one link (Section 2). Our question is narrower: *can an equal-capacity, direction-paired buffer recover useful elasticity while exposing only one shared read and one shared write service per pair and cycle?*

This paper makes four contributions:

- an equal-storage organization with three independent pairs per router, one private adaptive VC and one escape VC per direction, and four movable adaptive slots per pair;
- an alternating read schedule and a one-write admission guard that implement the shared bank as 1R1W;
- a conserved ownership-credit policy combining pressure, sticky ties, a three-credit cap, and idle-slot reclaim; and
- a 1,077-run evaluation that reports both the favorable region and the workloads where the fixed phase schedule is counterproductive.

2 Background and Motivation

A conventional four-VC input assigns four physical VC slots to each direction. Credit round-trip time determines how quickly a released downstream slot can be reused. With one-flit control traffic and an approximately 17-cycle credit loop at link latency $L = 8$, four slots impose an idealized zero-contention input-credit bound of

$$\lambda_{\text{credit}} \approx \frac{4}{17} = 0.235 \quad \text{flits/cycle.} \tag{1}$$

The exact saturation point can be lower because routing, output contention, and switch arbitration intervene. This bound is *per direction*, and that is the problem: under persistent $+X$ skew the E input exhausts its credits and stalls its upstream sender while the four W slots one crossbar port away stay idle. The router holds enough total storage for the offered load; it is partitioned along a boundary the traffic does not respect.

Elastic buffers are the classic remedy. DAMQ buffers manage one physical buffer as linked lists whose queues

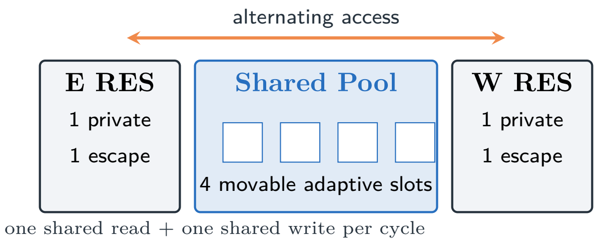

Figure 1: One direction pair. Every router contains three copies: E/W, N/S, and U/D. Capacity matches the private four-VC baseline.

grow with demand [12]; ViChaR dispenses VCs themselves from a unified buffer [10]; ElastiStore backs all VCs of a port with a shared slack buffer [11]. All share storage *within* one input port, and all pay in multiported storage, per-flit pointer management, or wider arbitration—costs that grow with the number of sharers. Production interconnects settled on a more conservative contract: a small private reservation per traffic class plus one shared, credit-managed pool (QPI/UPI's VN0/VNA [9], Omni-Path's dedicated and shared receive regions [4], and CMN-700's per-link pools [2]). But these pools share among the message classes of one physical link, so directional skew strands storage exactly as in the private baseline.

Tidal combines the two: it applies the reserved-plus-pooled contract *across* a router's two opposing inports—the axis skewed traffic actually unbalances—while constraining the shared bank up front to one read and one write per cycle, a budget conventional SRAM provides. Pairing exactly two sharers keeps the sharing hardware small, and admission moves by conserved ownership credits rather than per-flit pointers. Deadlock safety stays outside the elastic domain: every direction keeps a private deterministic escape VC, consistent with separating deadlock avoidance from adaptive resource use [6, 8].

3 Architecture

3.1 Direction-paired storage

Each router instantiates East/West, North/South, and Up/Down pairs. Figure 1 shows one pair. For each direction, res contains one private adaptive VC and one private escape VC. The pair shares four adaptive slots:

$$2(1_{\text{private}} + 1_{\text{escape}}) + 4_{\text{pool}} = 8 \text{ slots.} \tag{2}$$

This equals the eight slots in two four-VC private inputs, so performance comparisons preserve total storage.

Pool membership follows *true ingress*, not the selected output. A flit entering from X may occupy the X-pair Pool and later depart through Y or Z. Each pooled flit stores its ingress direction and upstream VC so that its credit returns to the correct sender.

3.2 Router-level on-chip organization

Figure 2 places one pair inside a router tile. The same structure is replicated for E/W, N/S, and U/D. The two incoming links retain independent res banks. Their adaptive arrivals also feed one shared Pool write arbiter. Section 3.4 details the pair's internal construction.

3.3 One-read, one-write service

The Pool accepts at most one read and one write service per pair, control virtual network, and cycle. Read phases are complementary:

Phase	Side 0 (E/N/U)	Side 1 (W/S/D)
Even	res	Pool
Odd	Pool	res

Thus E can inspect res while W inspects the Pool, and the roles reverse next cycle. Cycle parity structurally eliminates simultaneous cross-direction Pool reads. Ordinary VC and switch arbitration still select among eligible flits inside the active bank.

A per-pair write guard permits one adaptive Pool allocation each cycle. If both sides arrive, the second arrival may still use its private adaptive VC when that reserved credit is available; otherwise it waits. Escape storage never enters the shared bank and follows deterministic non-wrapping XYZ routing.

The port count must be stated precisely. A pair can perform two useful reads in a cycle: one from the selected side's res and one from the shared Pool. Only the latter consumes the Pool's single read port. A res write is likewise independent of the Pool write. For the present one-flit control traffic, Pool VC allocation and data arrival are one-to-one, so accepting at most one new Pool allocation per pair and cycle models a single Pool write service. A multi-flit extension would need a separate per-flit write schedule.

3.4 Physical construction of one pair

Figure 3 draws the blocks that the service contract above implies for one pair and control vnet.

Storage. The four Pool entries are flit-wide registers with a sidecar tag, built either as one central four-entry 1R1W register-file macro or as two two-entry halves colocated with each side's res bank and numbered globally; the Garnet model mirrors the second layout (Section 5.1). Each entry holds the 128-bit flit of the default Garnet link plus a seven-bit tag—true-ingress side (1 bit), credit inport (3 bits), and upstream logical VC (3 bits). The tag exists only because a slot may be borrowed by the opposite side; res entries need none.

Read path. The phase bit φ is the low bit of the router cycle counter. Per side it masks which bank's requests enter switch allocation; within the active Pool bank a masked round-robin pointer picks one eligible entry, and a 4:1 flit multiplexer steers it onto the winner's true-ingress lane. The crossbar keeps one port per direction and gains none.

Write path. Adaptive arrivals of both sides meet a two-request round-robin arbiter in front of the single write port; a token flop cleared every cycle enforces the single grant. The loser falls back to its private VC when that credit is free and otherwise waits upstream. res writes bypass the arbiter.

Control state and policy logic. Table 1 counts the

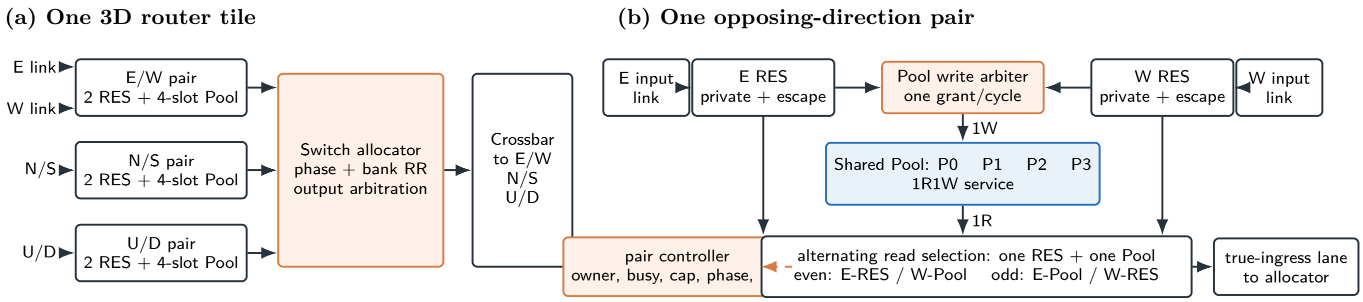

Figure 2: Proposed on-chip organization. (a) Three direction-pair controllers operate in parallel around the normal Garnet switch allocator and crossbar. (b) One pair exposes a single shared Pool read and write service. RES access remains separate; solid arrows are flit/request paths and the sole dashed arrow is controller state. The Pool result is placed on the flit's true-ingress lane.

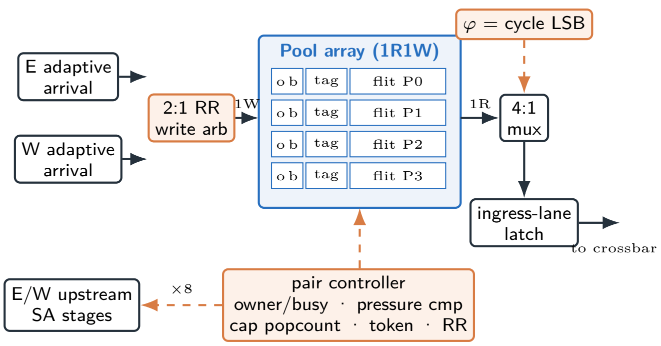

Figure 3: Construction of one pair's shared bank: a two-request arbiter in front of the single write port, a phase-gated 4:1 read multiplexer onto the winner's true-ingress crossbar lane, and a local pair controller whose owner/busy bits both upstream allocators observe over a narrow sideband. o, b = per-slot owner and busy bits.

Table 1: Per-pair, per-vnet control state.

State	Bits	Role
owner[0..3]	4	authorized side per slot
busy[0..3]	4	in-flight guard (from VC-valid)
write token	1	one POOL write per cycle
bank RR pointers	8	read selection per side/bank
policy RR bit	1	write-arbiter fairness
ingress tags	4×7	credit return for borrowed slots
total	46	< 5% of 1,024 data bits

pair controller's state: 18 control bits plus four seven-bit tags per pair and vnet, under 5% of the pair's 1,024 data bits. Pressure is a combinational count of each side's active RES VCs (range 0–2), so a release grant is a 2-bit compare; the cap check is a popcount of the four owner bits against three. All policy logic is local to the downstream pair controller.

Ownership observation. Owner and busy bits live at the downstream controller, but the claim decision runs in the upstream allocator, which therefore observes eight bits per pair and vnet over a one-way sideband (or an upstream-mirrored copy updated by the same events). The simulator treats this observation as combinational; Section 4.3 prices registering it.

Pipeline placement. Upstream, the owner/busy check

Table 2: The two credit planes in TIDAL.

Plane	State and path	Meaning
Occupancy	Garnet OutputUnit VC state; returned on the real CreditLink to the upstream port that sent the flit	The physical slot is free; prevents overflow
Ownership	One owner bit per Pool slot; selected by the downstream pair controller and observed by either opposing sender	This direction may allocate the idle slot

and the slot claim run inside output-VC selection at switch allocation and consume the write token in the claim cycle. Downstream, the phase mask gates input arbitration, switch traversal is unchanged, and the release-time pressure grant runs at tail departure together with the ordinary credit enqueue. The event-level Garnet model does not synthesize these blocks; it enforces their service limits and measures their utilization.

4 Pool-Credit Control

Each of the four shared slots has exactly one logical owner. For the E/W pair, ownership obeys

$$C_E + C_W = 4, \qquad 1 \le C_E, C_W \le 3. \tag{3}$$

An owned POOL credit authorizes the corresponding upstream direction to reserve an idle slot. Private and escape credits remain pinned.

4.1 Two credit planes

The implementation deliberately separates *buffer occupancy* from *Pool ownership*. They are both called credits informally, but they have different destinations and correctness roles:

Returning an occupancy credit never changes its physical path. Even when an E flit was placed in the W-hosted half of the POOL, stored metadata identifies the true E ingress link and the original upstream logical VC. The ordinary free credit is sent back on E's CreditLink. Independently, the downstream owner bit may stay with E or move to W.

A sender whose logical VC has become idle must still pass the owner-table check before reusing that Pool slot.

4.2 Allocation, release, and handoff

Figure 4 follows one one-flit packet. The upstream router first searches its private adaptive VC, then idle Pool slots it already owns, then idle peer-owned slots that it may reclaim under the cap. A successful Pool choice atomically sets `owner[slot]`, sets `busy[slot]`, and consumes the pair's write token for that cycle. The logical Pool VC ID travels with the flit. At the downstream router, that ID is mapped to one of four physical Pool entries and tagged with true ingress, credit input port, and upstream VC ID.

When the tail wins switch traversal, the downstream router first clears the busy bit and then evaluates the release policy. Table 3 gives the exact state transitions for the proposed pressure policy.

Pressure grant. When a pooled flit departs, its released credit is assigned to the side whose RES occupancy is larger. This local pressure signal approximates near-term need without adding a traffic predictor.

Sticky tie. Equal pressure preserves the previous owner. Stickiness avoids credit oscillation and favors the side likely to continue a burst.

Owner cap. A cap of three lets a hot side borrow one additional slot while guaranteeing one shared credit to its peer. Cap four is evaluated only as a full-elasticity upper bound.

On-demand reclaim. Release-only movement can freeze if an idle owner has no flit to release. A requester below its cap can therefore take an *idle* peer-owned slot immediately. Busy slots are never revoked. In the current simulator, this owner-table update adds no cycle; Section 8 treats this as an optimistic timing assumption.

The pressure value in the current code is the number of active private adaptive and escape VCs in RES. The just-released Pool slot is excluded. On equal pressure the old owner keeps the slot. If the preferred peer already owns three slots, the cap overrides pressure and the old owner keeps it.

4.3 Timing interpretation

The ordinary occupancy credit is not idealized: it is inserted into the original InputUnit's credit queue and scheduled on Garnet's CreditLink one router cycle later. Ownership handoff is the optimistic component. The simulator stores `owner[router][pair][vnet][slot]` at network scope, so the newly authorized upstream direction observes the new owner without a modeled sideband-wire or grant-register delay. A physical design needs a one-way owner-grant signal from the downstream pair controller to both opposing upstream allocators. Adding a registered one-cycle grant, or a parameterized delay, is necessary before drawing frequency-sensitive conclusions.

Blind owner round-robin is a useful control policy, but it is distinct from the round-robin arbitration inside the read scheduler. Rotating ownership without observing demand may park capacity at an inactive side.

5 Garnet Implementation

We implemented TIDAL in gem5 Garnet's router, input unit, VC allocator, and switch-allocation paths. The exact strict-1R1W snapshot is retained at `1R1W/overlay`; its provenance records implementation commit `dc43a290` and base commit `2eb6a4e`. Major additions are:

- physical-to-logical mapping for pair-global POOL slots;
- true-ingress and upstream-VC metadata for routing and credit return;
- complementary RES/POOL eligibility in switch allocation;
- a one-write-per-pair guard in downstream allocation;
- owner tables, pressure grants, cap enforcement, sticky ties, and idle-slot reclaim; and
- structural counters and X-biased/reversal synthetic traffic.

5.1 Physical and logical VC mapping

Each physical InputUnit still contains four control-vnet VC records. Offsets 0 and 3 are private; offsets 1 and 2 are that InputUnit's physical half of the pair Pool. An internal link exposes six *logical* IDs so either opposing sender can name all four shared slots:

On arrival, `dpPhysMapLogicalVC` translates the pair-global slot ID into a physical input direction and offset. A borrowed E flit may therefore be written into an entry resident in the W InputUnit. The VC records true direction, original credit input port, and original upstream VC. Routing uses the true direction, not the physical host, and switch traversal uses E's crossbar lane. On release, the saved fields steer the ordinary credit back to E.

5.2 Pipeline hooks

The C++ model carries exactly the control state of Table 1; statistics and ownership-wait timestamps are instrumentation, not proposed hardware. The relevant Garnet hooks are:

- `Router::dpPhysSelectAdaptiveVC`: private first, then owned Pool, then idle peer Pool under the cap;
- `InputUnit::wakeup`: logical-to-physical redirection and ingress metadata capture;
- `SwitchAllocator::arbitrate_dp_phys_inports`: complementary RES/Pool eligibility and per-bank round-robin selection;
- `InputUnit::increment_credit`: true-link credit return and, on the free signal, release-time owner selection; and
- `GarnetNetwork::dpPhysClaimSlot`: owner/busy update and the one-write-per-pair-per-cycle guard.

Experiments inject only virtual network 0 with one-flit control packets and one slot per control VC. Hence $L = 8$ means an eight-cycle link, not an eight-flit packet. Exported counters include POOL allocations, owned allocations, demand reclaims, release handoffs/keeps, owner

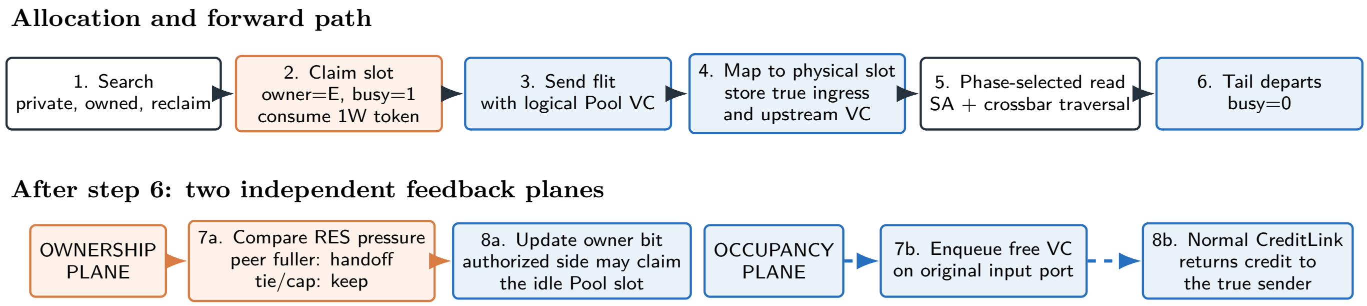

Figure 4: Pool-credit lifecycle. The ordinary free credit returns to the actual sender, whereas the released slot's ownership may be granted to either side. The two non-crossing feedback lanes are both triggered by step 6. The simulator models the owner-table change with no additional grant delay.

Table 3: Exact allocation and return rules for direction d and peer $\bar{d}$. An owner change is legal only for an idle slot.

Event	Guard / priority	Atomic state update	Result
Private allocation	Private adaptive VC is idle	Reserve the private VC; Pool state unchanged	No Pool port consumed
Owned Pool allocation	Private busy; Pool 1W token free; an idle slot has $\text{owner}[s] = d$	$\text{busy}[s] \leftarrow 1$; record current-cycle write token	Send using logical VC $1 + s$
Demand reclaim	No usable owned slot; 1W token free; peer-owned s is idle; $C_d <$ cap	$\text{owner}[s] \leftarrow d$; $\text{busy}[s] \leftarrow 1$; consume write token	Immediate reclaim; busy slots are never revoked
Second Pool request	Current-cycle write token already consumed	Do not allocate Pool; use private/escape only if independently legal	Otherwise request waits
Pool tail release	Selected Pool flit completes switch traversal	$\text{busy}[s] \leftarrow 0$; count RES occupancy on both sides	Evaluate pressure below
Pressure handoff	$\text{RESused}_{\bar{d}} > \text{RESused}_d$ and $C_{\bar{d}} <$ cap	$\text{owner}[s] \leftarrow \bar{d}$	Peer gains ownership; old sender still receives the normal free credit
Sticky keep	Tie, own side fuller, or peer at cap	$\text{owner}[s] \leftarrow d$	Preserve burst locality and the one-credit peer floor

Table 4: VC numbering for the four-physical-VC design.

Namespace	Offset(s)	Role
Physical InputUnit	0	private adaptive
Physical InputUnit	1, 2	resident Pool half
Physical InputUnit	3	private escape
Internal-link logical	0	private adaptive
Internal-link logical	1–4	pair-global Pool slots
Internal-link logical	5	private escape

migrations, write blocks, bank reads, per-direction activity, and peak borrowing.

6 Evaluation

6.1 Methodology

The primary baseline, PRIVATE-V4, dedicates four physical VCs to every input. All TIDAL arms use two RES slots per side plus four shared slots per pair, preserving eight slots. We compare blind owner round-robin, pressure cap three, and pressure cap four.

Tests use 4×4×4 or 8×4×4 3D torus networks, link latencies $L \in \{1, 8\}$, one-flit control packets, 10,000 simulated cycles for main long runs, and three seeds per reported point (the bias–rate map uses one). The 1,077-run main matrix contains 120 broad validation, 96 policy, 72 owner-cap, 192 reversal, and 171 X-biased/VC-scaling runs, plus 426 curve and map runs: 198 saturation-curve, 144 policy-curve, and 84 bias-grid points. A separate 48-run recovery-instrumentation rerun is not included in this total.

6.2 Structural validation

All 1,077 main runs completed without deadlock. Instrumented runs maintained credit conservation, never exceeded the owner cap, and satisfied

$$N_{\text{alloc}} = N_{\text{owned}} + N_{\text{reclaim}}, \tag{4}$$
$$N_{\text{migration}} = N_{\text{reclaim}} + N_{\text{handoff}}. \tag{5}$$

Per-direction allocation totals matched aggregate counts, and observed cross-direction POOL read conflicts were zero. Under cap three, peak borrowing was one slot, confirming that the peer retained its final credit.

6.3 Persistent X-direction skew

The 8×4×4 X-biased workload sends each packet three minimal X hops and varies the probability of choosing +X. It preserves spatially distributed destinations while

Table 5: Complete 1,077-run evaluation matrix. Rates are offered injection probabilities. Long-run rows report three-seed means and approximate 95% confidence intervals; validation is a single-seed smoke matrix.

CSV dataset	Purpose / traffic	Topology	L	Sweep points	Cycles × seeds	Runs
`validation_v4`	Sanity: uniform, X-opposite, tornado	4×4×4	1, 8	rate 0.1, 0.3, 0.5, 0.7, 0.9; four modes	3,000 × 1	120
`policy_v4`	Tornado policy comparison	4×4×4	1, 8	rate 0.1, 0.3, 0.5, 0.9; four modes	10,000 × 3	96
`cap_v4`	Tornado/uniform cap ablation	4×4×4	8	rate 0.3, 0.5, 0.9; caps 2, 3, 4 + private	10,000 × 3	72
`reversal_v4`	Alternating $+X/-X$ demand	4×4×4	1, 8	rate 0.3, 0.9; epoch 4, 16, 64, 256; four modes	10,000 × 3	192
`xbiased_bias_v4`	Persistent X-direction skew	8×4×4	8	rate 0.10; +X bias 50–100%; private/cap3/cap4	10,000 × 3	54
`xbiased_fullskew_v4`	Four-VC stability bracket	8×4×4	8	rate 0.08–0.12; 100% +X; three modes	10,000 × 3	45
`xbiased_vcscale`	Six/eight-VC scaling	8×4×4	8	rate 0.15, 0.18, 0.20, 0.25; six modes	10,000 × 3	72
`xbias_curve`	Saturation curves under X skew	8×4×4	8	rate 0.02–0.14, 11 points; bias 0.9, 1.0; private/cap3/cap4	10,000 × 3	198
`tornado_curve`	Tornado policy curves	4×4×4	8	rate 0.05–0.90, 12 points; four arms	10,000 × 3	144
`bias_heatmap`	Acceptance map	8×4×4	8	bias 0.5–1.0 × rate 0.02–0.14; private, cap3	10,000 × 1	84
					Total	**1,077**

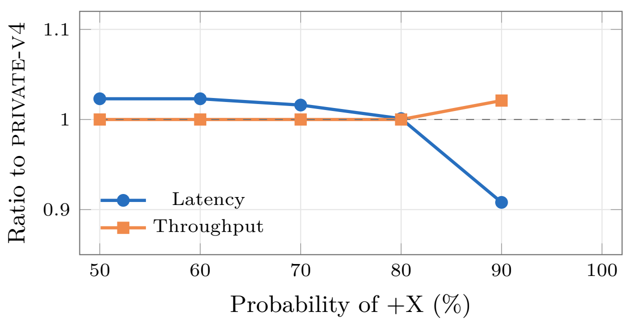

Figure 5: PRESSURE-CAP3 under X-direction skew at $L = 8$, injection 0.10. The fully skewed saturated point is excluded from the latency curve.

controlling directional imbalance. Figure 5 reports $L = 8$, injection 0.10, and four total VCs per input in the baseline.

At 50/50 and 60/40, TIDAL pays a 2.3% latency penalty because little cold-side capacity is stranded. The penalty falls to 1.6% at 70/30 and approximately breaks even at 80/20. At 90/10, latency falls by 9.2% while throughput rises by 2.1%. At 100% +X, the private baseline saturates, so a latency ratio is not a valid below-saturation comparison; the meaningful observation is 1.818 times delivered throughput during the measurement window.

Figure 6 traces the full saturation curves at 100% skew with 0.01-step resolution around each knee. Every arm rides flat near 47 cycles until its input credits run out, then collapses within one step: acceptance breaks between 0.08 and 0.09 for PRIVATE-V4, between 0.10 and 0.11 for cap three, and between 0.12 and 0.13 for cap four. Each knee sits within 2% of the idealized input-credit bound $C/(h\,T_{\mathrm{loop}})$ implied by Section 2—0.078, 0.098, and 0.118 for the $C = 4, 5, 6$ slots the arm's hot input may hold. Credit count, not switch or link bandwidth, is the binding resource here, and moving one or two idle peer credits raises the stable rate by exactly their share: 25% for cap three, 50% for cap four.

Figure 7 maps the operating region over the full bias–rate grid. Cap three's stable region strictly contains the baseline's: no cell is worse, the two maps coincide at balanced 50% bias, and for bias probabilities of 0.7 and above the acceptance frontier sits one 0.02 grid step further right—17–25% more stable offered rate precisely where skew persists. The map also bounds the mechanism's reach: below the private credit bound every organization is equivalent, so TIDAL's case rests entirely on the band between the private and pooled frontiers.

6.4 Tornado and policy comparison

On 4×4×4, torus tornado advances each packet by +1 in X, +1 in Y, and +1 in Z, making W, S, and D true inputs persistently hot. Figure 8 sweeps the offered rate at $L = 8$; Table 7 details the saturated 0.9 point.

Pressure cap three lowers latency by 4.85% without changing throughput materially because an output bottleneck limits tornado first. It captures about 83% of the latency benefit of cap four while preserving one peer credit. Only approximately 0.49% of POOL allocations use on-demand reclaim: stable ownership handles most demand, and reclaim resolves occasional mismatches. Owner round-robin loses 56.5% throughput because blind rotation can place credits at an inactive peer and has no demand-driven correction.

The rate sweep separates the failure modes. Below the output-bandwidth knee near 0.3, all demand-aware arms deliver identical throughput and the phase schedule costs the TIDAL arms a flat 1.6 cycles of latency over PRIVATE-V4; past the knee the order reverses and pressure rides 2.4 cycles below the baseline. Owner-RR is qualitatively different: its latency climbs from rate 0.1 onward, and at 0.25 its delivered throughput collapses to 0.068 and never recovers—misplaced credits, not bandwidth, are its binding resource.

Cap two prevents borrowing and is behaviorally equivalent to static ownership. Cap three recovers most of cap four's latency benefit while retaining one credit at the cold side. Cap four is slightly faster here but can temporarily leave the peer with no Pool ownership.

The event identities are visible numerically: $202{,}267 = 201{,}267 + 1{,}000$ allocations and $1{,}809 = 1{,}000 + 809$ migra-

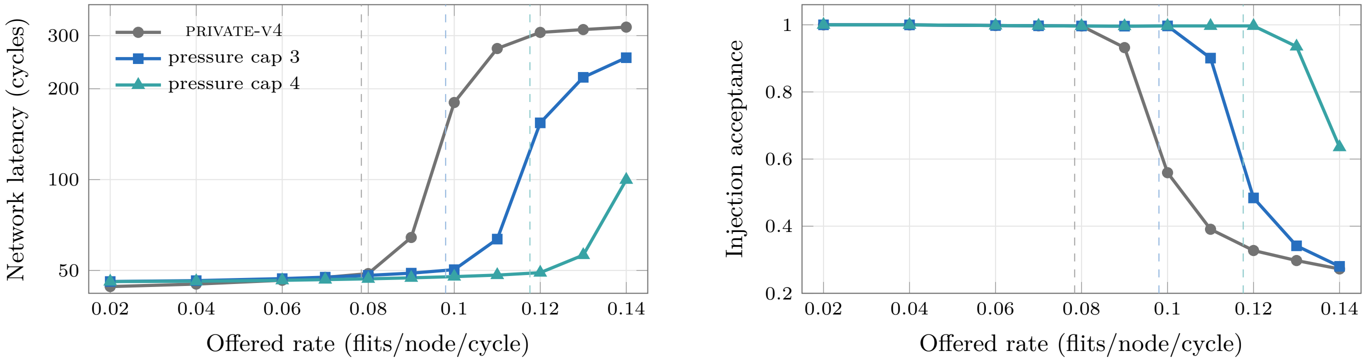

Figure 6: Saturation behavior under 100 % $+X$ skew at $L = 8$ (three-seed means). Dashed rules mark the input-credit bounds $C/(h\,T_{\mathrm{loop}}) = 0.078$, 0.098, 0.118 for the $C = 4$, 5, 6 slots each arm's hot input may hold (h=3 X hops, T_{loop}=17 cycles). Every measured knee lands on its bound.

Table 6: Detailed X-bias sweep at $L = 8$, offered rate 0.10. Values are mean ± approximate 95% confidence interval across three seeds; ratios are computed per seed before averaging. The dagger marks an above-saturation latency comparison.

$+X$	Private thr.	Cap3 thr.	Thr./base	Private net lat.	Cap3 net lat.	Cap3 lat./base	Cap4 lat./base
50%	.0496±.0004	.0496±.0004	1.000±.000	45.86±.02	46.94±.02	1.023±.000	1.018±.000
60%	.0496±.0004	.0496±.0004	1.000±.000	46.05±.06	47.09±.04	1.023±.001	1.015±.001
70%	.0496±.0004	.0496±.0004	1.000±.000	46.65±.08	47.39±.02	1.016±.002	1.005±.002
80%	.0496±.0004	.0496±.0004	1.000±.000	47.91±.15	47.93±.07	1.001±.002	.982±.002
90%	.0486±.0013	.0496±.0004	1.021±.028	53.94±4.68	48.78±.07	.908±.075	.880±.074
100%	.0273±.0008	.0496±.0004	1.818±.061	180.14±8.17	50.30±.24	.280±.012†	.265±.012†

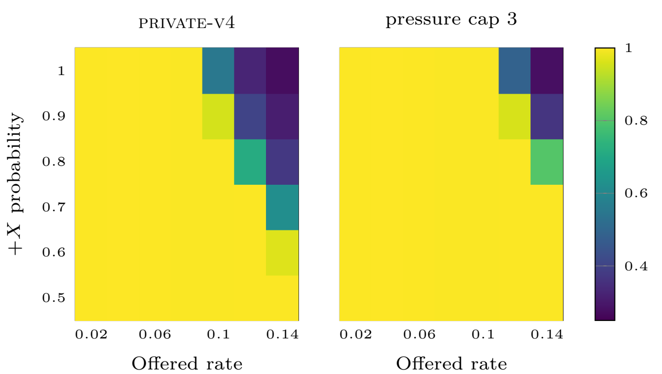

Figure 7: Injection acceptance over the bias–rate grid at $L = 8$ (single seed). Bright cells accept the offered load; dark cells are saturated. Cap three loses no cell, matches the baseline at balanced bias, and pushes the stable frontier one 0.02 step right wherever the skew probability is at least 0.7.

tions. Completed Pool reads also equal released keeps plus handoffs, 202,083 = 201,274 + 809; the allocation/read difference is the set of flits still resident at measurement end. Zero write-block counts mean the allocator never committed a second Pool write in the same pair-cycle in these runs; this validates the guard's behavior, not the timing or area of a fabricated 1W array.

6.5 Where 1R1W loses

The phase schedule is a real bandwidth constraint. Against PRIVATE-V4, uniform traffic at $L = 8$, injection 0.9 raises latency by approximately 3.2%. X-opposite traffic raises latency by 3–15% because both members of a pair compete for one shared read. Tornado at $L = 1$ raises latency by 25.1% because the credit loop is too short for additional capacity to repay the phase cost. Reversing the hot X direction every 16 cycles raises latency by 7.8% while ownership and old traffic drain in opposite directions.

The epoch-16 point is worst because ownership moves most often while the previous direction's flits are still draining. Very fast four-cycle reversal leaves little time to build an extreme ownership split; long epochs amortize each transition. This is direct evidence that credit policy must be evaluated as a dynamic controller, not only by its steady-state cap.

VC scaling supports the same interpretation. Near the measured knee, a six-VC organization reaches a 1.27 throughput ratio and a 0.56 latency ratio, extending the stable bracket from 0.12 to 0.15. At eight VCs, where private capacity already hides more of the credit loop, the ratios shrink to 1.08 and 0.83, extending 0.18 to 0.20.

6.6 Data provenance

The raw rows used above are tracked under `1R1W/overlay/lab/DP/physical_bound/`. Table 5 names each CSV exactly; the revision's curve and map rows live in `results_xbias_curve.csv`, `results_tornado_curve.csv`, and `results_bias_heatmap.csv`. `run_sweep_dpphys.py` records the gem5 command for every run, and `analyze_dpphys.py` performs paired-by-seed ratios and confidence-interval aggregation. The archived overlay should be reconstructed in a temporary worktree from base commit `2eb6a4e`; it should not be copied over the repository's current working source.

7 Hardware Interpretation

The evaluated comparison establishes equal buffer capacity and a one-read, one-write POOL service limit. This is a plausible manufacturing advantage over an ideal shared

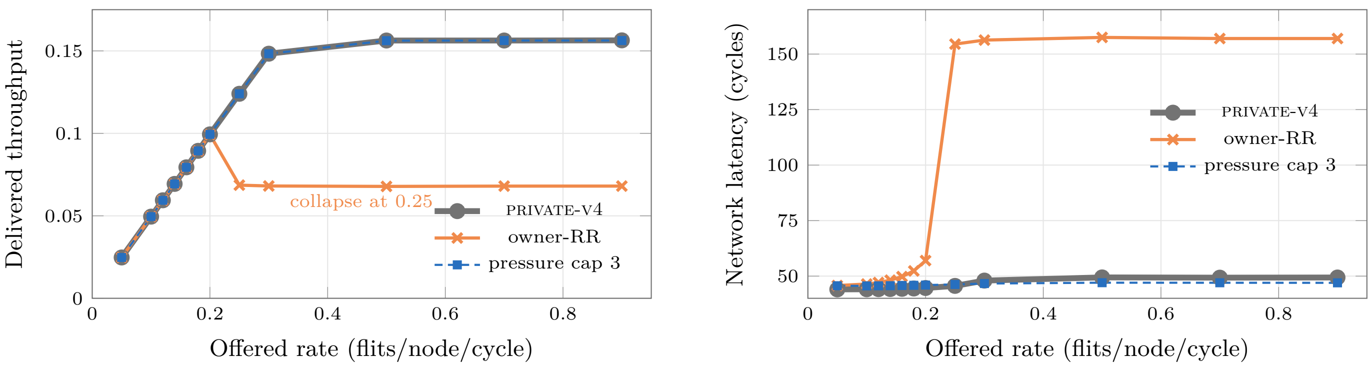

Figure 8: Tornado sweep at $L = 8$ (three-seed means; cap 4 overlays cap 3 and is omitted). Left: pressure tracks the private baseline to saturation, while blind owner rotation collapses at rate 0.25 and never recovers. Right: below saturation the phase schedule costs pressure a flat 1.6 cycles; past the output knee near 0.3 the order reverses and pressure finishes 2.4 cycles below the baseline.

Table 7: Tornado policy comparison at offered rate 0.9. Values are mean ± approximate 95% confidence interval over three seeds. Pool, reclaim, and handoff columns are percentages of completed control events.

L	Mode	Throughput	Thr./base	Net latency	Lat./base	Pool service	Reclaim	Handoff
1	PRIVATE-V4	.4489±.0003	1.000	10.46±.00	1.000	–	–	–
1	Owner-RR	.3138±.0013	.699±.003	32.08±.16	3.067±.016	25.37	.00	.11
1	Pressure cap3	.4488±.0003	1.000±.000	13.09±.01	1.251±.001	59.11	.13	.09
8	PRIVATE-V4	.1564±.0000	1.000	49.37±.07	1.000	–	–	–
8	Owner-RR	.0680±.0004	.435±.003	157.03±1.16	3.181±.022	27.67	.00	.44
8	Pressure cap3	.1563±.0000	1.000±.000	46.97±.03	.952±.002	66.38	.49	.40

Table 8: Owner-cap ablation for tornado, $L = 8$, rate 0.9.

Mode	Lat.	Lat./base	Reclaim	Borrow peak
PRIVATE-V4	49.37	1.000	–	0
Cap 2	58.25	1.180	0.00%	0
Cap 3	46.97	.952	.49%	1
Cap 4	46.47	.941	.19%	2

Table 9: Credit/control event counts for pressure cap 3 under tornado, $L = 8$, rate 0.9. Counts are rounded three-seed means per 10,000-cycle run.

Counter	Mean	Counter	Mean
Pool allocations	202,267	Owned allocations	201,267
Demand reclaims	1,000	Owner migrations	1,809
Release handoffs	809	Release keeps	201,274
Pool reads	202,083	RES reads	102,347
Pool write blocks	0	Pool read conflicts	0
Borrowed peak	1	Deadlocks	0

bank that must serve both directions simultaneously, but it is not yet an area or energy result. TIDAL adds the control state, selectors, arbiters, and ownership sideband of Section 3.4; the state itself is under 5% of the pair's data bits, but the selectors and the ownership update path can be timing-critical. We therefore claim reduced shared-port demand, not zero hardware cost. RTL synthesis against private and dual-access pool implementations is required to quantify area, energy, and frequency.

This comparison explains both the intended cost saving and the performance losses. Relative to an unrestricted 2R2W shared pool, TIDAL removes one shared read and one shared write service, reducing the hardest shared-port requirement. Relative to private banks, however, it

Table 10: Representative pressure-cap3 losses. Ratios are against equal-storage PRIVATE-V4. The X-opposite row is a 3,000-cycle single-seed smoke point; the others are 10,000-cycle, three-seed results.

Traffic / condition	L	Thr./base	Lat./base
Uniform, rate .9	8	1.001	1.032
X-opposite, rate .9	8	.999	1.154
Tornado, rate .9	1	1.000	1.251
X reversal, epoch 16	8	.999	1.078

Table 11: Direction-reversal sensitivity at $L = 8$, rate 0.9. Counts and absolute values are three-seed means.

Epoch	Base thr.	Cap3 thr.	Base lat.	Cap3 lat.	Migrations
4	.1575	.1575	27.37	27.97	5,335
16	.1560	.1559	28.49	30.72	16,707
64	.1549	.1558	27.77	28.60	7,649
256	.1565	.1567	26.63	27.26	2,428

adds a central selection and ownership path. Whether the smaller Pool port count outweighs those selectors depends on entry width, SRAM/register-file implementation, wire length, and target frequency. Garnet cannot answer that manufacturing question. The ordinary router allocator and crossbar remain present in all three organizations [7].

8 Limitations and Next Steps

The present evidence is preliminary and simulator-level. First, demand reclaim updates ownership with zero additional cycles; a one-cycle and parameterized grant path should measure timing sensitivity. Second, finer 0.005 injection steps, longer drain-aware runs, and at least five seeds are needed around every saturation knee. Third, a directed simultaneous E/W-write microbenchmark should stress the write guard. Fourth, multi-flit packets require an explicit packet/VC ownership-retention rule. Finally, RTL synthesis should compare private buffers, 1R1W Tidal, and an idealized dual-access pool.

9 Related Work

Virtual-channel flow control and its buffer/throughput tradeoffs were formalized by Dally [5]. Deadlock-free routing can be constructed through channel-dependency restrictions [6] or broader adaptive conditions [8]; Tidal preserves a private deterministic escape resource rather than changing those guarantees. Garnet provides the detailed router and credit model used here [1], inside the gem5 simulation infrastructure [3]. Shared-buffer organizations and industrial credit pools are positioned in Section 2; relative to both lines, our contribution is the combination of direction-paired equal-capacity storage spanning two opposing inports, a structurally 1R1W shared bank, and demand-aware ownership credits.

10 Division of Labor

The project was completed with equal overall contribution: Changxun Pan 50% and Liming Wang 50%. Both contributors participated in architecture discussion, debugging, correctness review, experiment review, interpretation, writing, slide production, and presentation preparation.

Liming Wang proposed the original direction-paired pooling concept, implemented and evaluated the earlier 1R2W organization, expanded traffic and evaluation coverage, reviewed the 1R1W and credit-policy behavior, and co-analyzed and co-authored the final deliverables.

Changxun Pan co-developed the design and finalized its complementary 1R1W phase schedule, designed pressure, sticky-tie, cap-three, and reclaim control, implemented the final mechanism and instrumentation, ran and cross-checked major 1R1W experiments, and co-analyzed and co-authored the final deliverables.

11 Conclusion

Tidal shows that buffer elasticity does not require an unconstrained global pool. Pairing opposite directions and preserving private escape capacity lets a four-slot adaptive pool move storage toward sustained pressure while retaining equal total capacity. A complementary schedule enforces one shared read and one shared write per cycle. The design is effective under persistent directional skew and long credit loops, but balanced demand, short links, and fast reversals expose its serialized-access cost. Pressure-aware credit movement is essential: blind ownership rotation is substantially worse. The next decisive step is to add grant delay and synthesize the control and storage organization so that performance gains can be judged against measured area, energy, and timing.

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